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Track: Digital Inclusion
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UMATE

Practise the metro in VR. Then ride it for real.

UMATE helps young people with autism, intellectual disability and other cognitive differences learn to travel on the Baku Metro and pay with BakıKart on their own. The learner puts an ordinary smartphone into a low-cost VR box and rehearses their own trip: top up the card, pass the turnstile, choose the right platform, get off at the right station. A calm AI voice buddy is beside them. A trainer watches every step live on a laptop. When the rehearsal is over, the same phone leaves the box and becomes the learner's guide on the real trip. On stage, one judge wears the headset while a second judge picks a problem from the laptop, for example "not enough balance". The first judge has to solve it in VR, and the audience sees both views.

Who

Young people with cognitive differences, their trainers (rehabilitation centres, inclusive schools) and families.

What

A phone and VR-box rehearsal of the Baku Metro and BakıKart payments, built from real station photos.

How

Look-to-select, a voice AI buddy, step-by-step support that fades over time, and a live trainer panel.

Why it matters

Without travel training, people depend on others or on costly special transport. Practising safely costs less and builds confidence before the real trip.

Contents

1. The problem
2. The solution and how it works
3. The prototype we are building
4. Research and competitors
5. Data
6. Where AI is used
7. Business model and roadmap
8. Hard questions and answers
9. Sources

Labels used in this document: **Assumption** means not yet verified, and **Plan** means not built yet.

1 The problem

Many young people with autism or intellectual disability can learn to travel alone. What they lack is safe, repeated practice, and digital payment makes the first trip even harder.

How many people this affects

Fact	Figure	Source
Autism among 8-year-olds in the USA (2022 data)	1 in 31 children (3.2%). About 40% of them also have an intellectual disability (39.6% in report summaries).	CDC ADDM Network, MMWR 2025
People with disabilities in Azerbaijan	About 530,000 to 620,000 people (5.6–6.4% of the population), depending on the year and the source. This total covers all disabilities, not only cognitive ones.	UNDP Azerbaijan 2019; World Bank 2022; Labour Ministry figures reported in 2018
Cost of depending on special transport	An ADA paratransit trip cost **$29.30**, against **$8.15** for a regular bus or train trip, about 3.5 times more (2010 data).	US GAO, GAO-13-17
Value of travel training	Net savings of **$233,000** a year at RTC Washoe (Reno) and about **$260,000** a year at Intercity Transit (Olympia).	TCRP / National Academies
Public funding exists elsewhere	Australia's NDIS funds "training to use public transport" as a disability support.	NDIS

Why the Baku Metro is a digital barrier

- Entry is only through turnstiles that read a BakıKart (plastic or paper), a QR ticket from the BakıKart app, or an NFC bank card. Validation happens at entry.
- Cards are topped up at kiosks with several screens and languages (Azerbaijani, Russian, English). A wrong choice, a card that is not read, or "not enough balance" in a crowd is exactly the moment many learners freeze.
- Choosing the platform means reading a sign "towards" a terminus station, not towards your destination. This is abstract and hard for many learners.

What happens today

Travel training is done one-to-one in the real metro: a trainer goes with the learner, often for many trips. It works, but it is expensive, stressful in crowds, hard to repeat and hard to measure. Existing VR tools need standalone headsets and free floor space (details in section 4).

2 The solution and how it works

UMATE turns any smartphone and a cheap VR box into a safe rehearsal of the learner's own metro trip. An AI buddy stays beside the learner and a trainer watches live.

The trip: 4 steps, each with a problem the learner practises

#	Step	What the learner does	Problem practised
1	Top up at the BakıKart kiosk	Looks at the kiosk, places the card, chooses an amount, checks the new balance.	Card not read
2	Pass the turnstile	Taps the card on the reader and waits for the green signal.	Not enough balance, so go back and top up
3	Choose the platform	Reads the two direction signs and picks "towards İçərişəhər" to reach 28 May.	Delay or platform change announced
4	Ride and get off Plan	Listens to announcements, counts stops, gets off at the right station.	Crowd and noise

How the learner interacts, without a controller

- **Look around:** the phone's gyroscope turns the view 360°. The learner sits on a swivel chair.
- **Choose:** looking at an object for 1.5 seconds selects it, while a ring fills up. A tap on the screen, the VR box button or a Bluetooth remote also works.
- **Walk:** looking at a target on the floor moves the learner there with a short fade. There is no smooth movement, which is the main cause of motion sickness.
- **Talk to the buddy:** looking at the glowing orb opens the microphone. In a noisy room it does not listen otherwise.

Support that fades, and mistakes that are allowed

- If the learner is stuck, help comes in steps: first the buddy gives a short spoken hint, then an arrow appears, then the right object is highlighted.
- Mistakes are allowed. The buddy explains calmly and the learner tries again. The panel records each mistake.
- **Asking for help counts as success**, as it does in special education practice.

The trainer panel and the stage demo

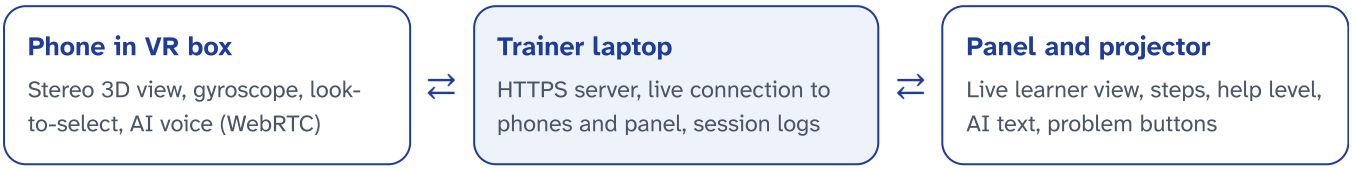
- The laptop shows a live view of what the learner sees (no video is streamed; the laptop renders the same scene from the learner's head position), plus the step timeline, the level of help used, the AI conversation as text and the virtual card balance.
- The trainer, or a second judge, can trigger a problem: *not enough balance, card not read, train delay or platform change, crowd and noise*.
- After the rehearsal, the phone leaves the box and switches to **guide mode**: the same steps as simple picture and voice cards, plus an "I need help" button.

> **Honest boundary.** Success in VR is evidence of practice, not a certificate that the learner is ready to travel

alone. The trainer decides when to try the real trip.

3 The prototype we are building

A working web prototype: nothing to install from an app store. It opens on Android and iPhone from a link or QR code, and runs on the trainer's laptop network.



Cloud services: OpenAI Realtime (voice buddy) and Gemini (trainer reports). If the internet drops, only the AI voice stops; VR, the panel and the logs keep working.

Status

Status	What
Done	Phase 0, the base: • local HTTPS server with its own certificate; • stereo VR with gyroscope, look-to-select and teleport; • live trainer panel that mirrors the learner's view; • mini flow: top up, turnstile with the "not enough balance" branch, direction sign; • AI buddy voice code; • automated checks (CI) and a public GitHub repository where every change needs approval from 3 teammates.
In progress	• Realistic 360° panoramas generated from real Baku Metro photos (the Koroğlu platform and the Bakmil turnstiles); • Baku-Metro-style signs (blue boards, red "M", official line colours, station names in Azerbaijani); • a voice-reactive aurora orb as the buddy (no cartoon face); • the BakıKart kiosk screen flow; • a font designed for low-vision readers (Atkinson Hyperlegible Next).
Not tested yet	The AI buddy voice on real phones, and the full flow on Android and iPhone inside the VR box.

Plan for the camp

Phase	Content
1	Scenario engine and data: • metro stations from OpenStreetMap, so the route steps and direction are built automatically; • a rules file (fare, top-up) that can be edited live; • automatic tests: the same input always gives the same result, and a failed payment never changes the balance.

2	All 4 steps in VR, with the effects of the four problems.
3	Panel complete: problem buttons, live rule changes, session log.
4	AI buddy connected to the scenario engine: • it receives the facts of the current step; • it gives hints on its own when the learner is stuck and calms them after mistakes; • pre-recorded lines are used when there is no internet.
5	Guide mode, AI trainer report, and sample progress charts clearly labelled "SAMPLE".
6	Reliability and rehearsal: • 10 full runs in a row on both phones; • a 60-second backup video; • 3-minute and 7-minute demo scripts.

Technology: three.js 0.187 (3D), Node.js 24, WebSocket, OpenAI Realtime over WebRTC, Gemini, GitHub Actions. Code: github.com/metinmehdi/UMATE

4 Research and competitors

VR travel training looks promising, but its transfer to real life is the weakest part of the evidence. We therefore design UMATE around the bridge to the real trip.

What the research says

Study	Finding	Limitation
Simões et al., 2018, *JMIR Serious Games*	A VR bus-travel game for autistic participants (n=10, 1–3 sessions). Knowledge of bus riding rose significantly, in-game success was high (93.8% reported), and electrodermal activity, a marker of anxiety, fell inside the virtual bus.	Small proof of concept with no autistic control group.
Michalski et al., 2021, *Frontiers in Psychology*	A systematic review of 8 studies on vocational training in virtual environments for neurodevelopmental disorders. Skills can transfer to real settings.	Few studies, and about work skills rather than travel.
Education Sciences, 2025 (MDPI)	A systematic review of VR for adaptive behaviour in intellectual disability. Learning must extend into real community settings for transfer.	Transfer is not automatic.
Schmidt et al., 2021, *Frontiers in Virtual Reality*	Risks for autistic users: cybersickness, falls, anxiety, sensory overload. The recommended answer is gradual acclimation.	We follow this in our design.

Competitors

Product	What it does	Gap we address
Floreo (USA)	VR lessons for neurodivergent learners, on Meta Quest 2 or iPhone. A Kennedy Krieger pilot with 4 young adults practises waiting for the bus and crossing streets.	General lessons rather than the learner's own route; no local payment system. Pilot outcomes are not published.
Changefied Virtual OV (Netherlands)	Public-transport VR training for neurodivergent users aged 12+: route planning, transfers, crowds, delays, asking for help. Personalised, with pre/post tests. 4,000+ people trained, 4.2/5 rating.	Needs a Quest, Pico or Vive headset and a 2×2 m space. It states that its effect is not measured. Using your own route is not mentioned.
Academic prototypes	Campus bus replicas and bus simulators (USA, Switzerland).	Research only, not products.

How UMATE differs

	Typical VR training	UMATE
Hardware	Standalone headset (hundreds of dollars) and free floor space	Any smartphone and a low-cost VR box, used seated
Route	Generic city	The learner's real Baku Metro route, built from real photos and open data
Digital payment	Rarely included	BakıKart top-up and turnstile, including failure cases
Bridge to real life	Training ends in the headset	The same phone becomes the real-trip guide with the same steps
Live problems	Scripted	The trainer (or a judge) triggers problems in real time
Measurement	Often not measured	Every step logged; real-trip outcomes recorded in the pilot Plan

5 Data

We only use data that changes a decision: how the route is built, how the scene looks, or what the trainer does next.

Data we use

Data	Source	Used for
Baku Metro stations and lines	OpenStreetMap Plan, phase 1	Builds the route steps and the correct direction automatically for any two stations

Official line colours	Baku Metro line module (Wikipedia): Red #EE1D23, Green #00AB5B, Purple #8E177C	Authentic signs
Daily passenger trips by station	opendata.az: Bakı Metropoliteni	Realistic crowd levels per station **Plan**
Payment rules (fare, top-up)	A rules file in the app. The demo fare is 0.60 AZN **Assumption: check the current tariff**	Live-editable rules: change the fare and the scenario behaves differently at once
Open transit data worldwide	Mobility Database: 6,000+ GTFS feeds in 99+ countries	Adding new cities without new code
Reference photos	Wikimedia Commons (Bakmil turnstiles, Koroğlu station)	Realistic 360° panoramas

Data each session creates

A time-stamped event log stored on the trainer's laptop. It records:

- session start;
- step reached;
- what the learner looked at and chose;
- mistakes;
- help shown (level and reason);
- problems triggered and solved;
- card transactions (balance before and after, success or failure);
- the AI conversation as **text only**;
- head direction 5 times a second;
- a summary of each step (time, mistakes, highest help level, whether help was asked for).

All numbers on the panel are calculated from this log.

Real versus sample data

The live session on stage is real. Charts showing progress over several weeks use a **synthetic group of learners, clearly labelled "SAMPLE"**, until pilot data exists. We never present sample data as evidence that UMATE works.

Privacy

- No login: learners appear as nicknames.
- Data stays on the trainer's laptop. No audio is recorded or stored.
- During a session, the learner's voice is processed by the AI provider to produce the answer.
- A real product needs guardian consent, a data-processing agreement, and an option to turn the voice off.

6 Where AI is used

AI does what one trainer cannot do for many learners at once: it is present every minute, notices when someone is stuck, and summarises progress.

Where	Model or tool	What it does	Safeguard
Voice buddy	OpenAI Realtime (gpt-realtime-2.1) over WebRTC	Answers questions, gives short hints, calms the learner with a breathing exercise. It listens only when looked at.	It speaks only from facts the scenario engine sends. It may not invent stations or prices. Sentences are short and calm. Without internet it falls back to pre-recorded lines.
Timing of help	Scenario engine (rules)	Increases help after waiting (spoken hint, then arrow, then highlight).	Deterministic and logged.
Trainer report	Gemini	Turns the session log into a short summary and a suggestion for the next lesson.	Template report when offline; the trainer reviews it.
Visual content	Codex image generation	Creates 360° station panoramas from real reference photos.	Reviewed by the team; no people and no fake text.

Our own AI, on the roadmap

- **Readiness prediction model** Plan:
  - In the pilot, trainers record whether the real trip succeeded.
  - A model learns which VR behaviours predict real independence.
  - This turns our weakest point (VR is not real life) into a measurable advantage, using data nobody else collects.
  - It creates a loop: more sessions give better predictions, which earn more trust, which bring more centres and more sessions.
- Later: Azerbaijani speech recognition for atypical speech, and a small AI that runs on the phone itself for privacy and offline use.

We do not claim to have trained our own model yet. Today we use existing AI services carefully, and we design the data collection that a future model needs.

7 Business model and roadmap

Centres pay, families don't.

Question	Answer
Who pays	Rehabilitation centres and inclusive schools, through an annual licence. State social programmes can fund it, as travel training is funded elsewhere (for example the NDIS in Australia).

Who uses it free	Families: the guide mode on the learner's own phone.
Why they pay	One-to-one real-world trips are the costly bottleneck. VR rehearsal lets one trainer run more sessions safely and gives progress reports for funders.
Cost advantage	Any smartphone and a low-cost VR box, instead of a standalone headset (hundreds of dollars) plus a 2×2 m room.
Pricing	Proposal A free 3-month pilot, then an annual centre licence, plus paid city packs. Prices will be set together with pilot centres; we give no numbers yet.
Our costs	AI voice minutes (paid per use). Hosting is minimal because the system runs on the trainer's laptop.
Scaling	A new city is a data pack (open GTFS/OSM data, photos, panoramas). A new language is a text file.

Roadmap Plan

When	Goal
Camp, Nov 2026	A working demo: 4 steps, AI buddy, trainer panel, problems triggered live, guide mode.
Next 3 months	Pilot with 1–2 centres in Baku; Azerbaijani language pack; pre/post knowledge checks; recording real-trip outcomes.
6–12 months	All metro lines and buses; first version of the readiness model; a second city.
Later	Recognition of atypical speech; AI that runs on the phone; more everyday skills (shops, bank card, appointments).

8 Hard questions and answers

These are the questions we expect from judges, with answers that do not overclaim.

1. VR is not real life. Will the skills transfer?

We agree that transfer is the weakest point in the research. That is why we use the learner's own real route, use real station photos, keep the same steps in guide mode on the real trip, and let trainers record real-trip outcomes. VR success is never treated as a readiness certificate.

2. Why not just use Changefied or Floreo?

They prove there is demand. Changefied needs a standalone headset and a 2×2 m space, and Floreo offers general lessons. UMATE runs on any phone with a VR box, trains BakıKart payment and its failure cases on the local route, connects to the real trip through guide mode, and lets a trainer trigger problems live.

3. What about motion sickness and sensory overload?

There is no smooth movement (only teleport), the learner sits, and first sessions last 2–3 minutes. Visuals are calm, the trainer controls the noise level, and the buddy can lead a breathing exercise. This follows the gradual-acclimation advice of Schmidt et al. (2021).

4. Isn't sending minors' voices to OpenAI a privacy problem?

On stage, adults use the demo. The microphone opens only when the learner looks at the buddy. No audio is stored, only text on the trainer's laptop. A real product would require guardian consent, offer a mode without voice (pre-recorded lines) and, later, AI running on the phone.

5. Why is the interface in English if the users are Azerbaijani?

English was chosen for the camp pitch. All interface text is in a single language file, so an Azerbaijani pack is the next step. Station names already appear in Azerbaijani, as on real signs. The quality of the AI voice in Azerbaijani has to be tested first.

6. What if there is no internet?

VR, the panel and the logs run on the laptop's local network. Only the AI voice needs the internet, and without it the buddy switches to pre-recorded lines.

7. How will you prove it works?

Today we measure behaviour in VR: time, mistakes, help level and help requests. In the pilot we add knowledge checks before and after, and trainer-recorded outcomes of real trips. We do not claim proven effectiveness yet.

8. Is the dashboard data real?

The live stage session is real. Multi-week trend charts use synthetic data that is clearly labelled "SAMPLE" until we have pilot data.

9. Can the AI give wrong directions?

The buddy receives the facts from the scenario engine and is told never to invent stations or prices. Whether an answer is right is decided by the engine, not by the AI. The trainer sees every transcript.

10. Who pays, and why would they?

Centres and schools pay, because UMATE saves trainer time and provides progress reports. Families use guide mode for free. Prices will be validated in the pilot.

11. Is a cheap VR box good enough?

For seated rehearsal, head rotation is enough. We designed for it: look-to-select and teleport. UMATE also works without a box by moving the phone in the hand. Testing on Android and iPhone is under way.

12. Did you really build this yourselves?

The code is public on GitHub with its full history, and every change was approved by three teammates. We openly used AI coding assistants. The architecture, the product decisions and the testing are ours.

13. How do you scale to other cities?

Routes come from open transit data (6,000+ GTFS feeds worldwide) and OpenStreetMap, the rules from a file, and the scenes from local photos. Language is a text file. None of this requires new code.

14. What if a learner panics in VR?

The learner can look away or take the phone out at any time. The trainer has a "calm down" button, the buddy guides a slow breath, and the session can stop at once. The trainer stays next to the learner.

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